HOUSE PRICE PREDICTION

Machine Learning Project Proposal

1. Project Overview

The aim of this project is to build a machine learning model that estimates the sale price of a house from details such as living area, number of bedrooms and bathrooms, overall quality, and other property features. The work will be completed in Python, with the training and experiments carried out in Kaggle Notebooks.

2. Dataset

We will use the well-known **House Prices – Advanced Regression Techniques** dataset from Kaggle. It contains information about houses in Ames, Iowa, and is commonly used for practicing regression and feature engineering.

Dataset: Kaggle House Prices – Advanced Regression Techniques

3. Tools and Libraries

The project will be developed in **Python**. Pandas and NumPy will be used to handle the data, Matplotlib and Seaborn will help with charts and analysis, and Scikit-learn will be used for preprocessing, model training, and evaluation. We may also test XGBoost or another boosting model to compare its results. Joblib will be used to save the trained model.

4. Project Steps

• Load the dataset in Kaggle and inspect its columns and missing values.

• Clean the data and prepare numerical and categorical features for training.

• Explore the data with charts to understand which features are related to house prices.

• Train and compare regression models such as Linear Regression, Random Forest, and Gradient Boosting.

• Evaluate the models using suitable measures such as RMSE and MAE, then save the selected model.

DEPLOYMENT AND EXPECTED OUTCOME

How the model will be made available for testing

5. Online Demo

Once the model is trained and tested, it will be deployed on **Hugging Face Spaces**. A small web interface will be created with **Gradio**, where users can enter house details and receive a predicted price. This will make the project easy to try without installing Python or running the notebook.

6. Deployment Plan

• Save the final model and the preprocessing steps used during training.

• Create a Gradio interface with input fields for the selected house features.

• Upload the application files to Hugging Face Spaces and test the public demo.

• Share the direct demo link so it can be opened in a normal web browser.

7. Expected Deliverables

• A Kaggle Notebook showing data analysis, preprocessing, training, and model evaluation.

• A saved machine learning model with the required preprocessing pipeline.

• A public Hugging Face demo where users can test house price predictions.

• Basic instructions explaining how to use the demo.

8. Expected Result

The final result will be a working house price prediction application supported by a complete Kaggle notebook. The model's performance will depend on the data and the results of testing, so the best-performing model will be selected after comparing the evaluation scores.

Important Note

The selected dataset is based on houses in Ames, Iowa. The model is intended for properties similar to those in the dataset and should not be used to estimate current house prices in Pakistan without suitable local data and further training.